Which Visual Representations Drive The Emergence of Numerosity in a Manipulation Task

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Abstract

Number sense, the ability to estimate the number of objects in a visual scene, appears across species and developmental stages. Here we demonstrate that explicit numerosity encoding emerge when pre-trained object classification networks are applied within an ecologically grounded manipulation recognition task, despite the absence of explicit number labels. Using photorealistic stimuli with controlled scene statistics, a siamese architecture trained to discriminate three manipulation actions (Take / Put / Shake) produced one-dimensional embeddings ordered by numerosity. Embedding quality correlated strongly with task accuracy (R = 0.92). Feature sets extracted from intermediate network layers yielded superior task performance and stronger numerosity encoding than feature sets from early or late layers. Post-hoc analyses indicate that the learned numerosity responses exhibit hallmark properties reported in primate studies, including increasing tuning bandwidth and positive skew. These results indicate that a primate-like number sense emerges from common visual interactions and is driven primarily by intermediate visual representations.

Keywords: Number-sense, Emergent Representations

1 Main

Glancing out of a window, you observe a group of friends socializing. Despite not counting their number intentionally, you are instantly aware that the group consists of four members. This is an example of visual number sense, a phenomenon which is observed even in preverbal children and non-human primates [1–3].

The ubiquitousness of number sense raises several interesting questions: How does numerosity emerge despite the absence of explicit number labels? Is this number sense a byproduct of other visual tasks? What cognitive mechanism drive the emergence of numerosity in visual systems?

Expanding on the rich body of literature focused on the primate number sense [1–4], researcher have also leveraged computational methods to investigate the emergence of number sense in artificial visual systems. For instance, models trained to directly predict numerosity from images learned representations that exhibited various human like characteristics [4, 5]. Recently, researcher demonstrated the emergence of similar human-like numerosity representations also occurs in deep neural networks (DNNs) trained with an unsupervised auto-encoding function, rather than supervised numerosity prediction [6]. Moreover, the developments of the numerosity discrimination ability in these DNNs followed trajectories similar to those observed in human infants, especially when using training stimuli that shares statistical attributes approximating those found in natural scenes. Similarly, Kondapaneni et. al. trained DNNs using an ecologically feasible 'manipulation recognition' task, resulting in learned representations that explicitly encode numerosity [7]. Their analysis revealed that the trained models embedded images into a one dimensional line that follows the order of number of objects in the stimuli. These recent works focused numerosity representations in models trained to perform 'ecologically feasible' visual tasks that do not require explicit number labels. However, their use of artificial stimuli consisting of grayscale rectangles against a dark background is potentially problematic as numerosity could be confounded with various visual statistics. To overcome this, Chapalain et. al. generated controlled photorealistic natural stimuli to disentangle numerosity from visual statistics [8]. Specifically, the authors found that pretrained object recognition models predicted numerosity above and beyond any tested visual statistic, while the same models without pretraining only encoded low-level visual statistics, highlighting the importance of pretraining.

Here we leverage these computational methods to explore how model properties impact the numerosity encoding that emerges in their learned representations. Utilizing the ecologically feasible framework proposed by [7] with photorealistic natural stimuli [8] and real world visual statistics [6] we explore which DNN characteristics could drive the emergence of a primate-like number sense. Moreover, we analyze the effect of pretraining on different visual tasks, and fine-tuning on stimuli with various degrees of visual complexity. Specifically, we assessed whether conclusions from previous research hold when using ecologically feasible stimuli and learning framework. Additionally, we investigated the impact of using model representations extracted from different DNN layers and models trained to perform various vision tasks.

Our results demonstrate that using a pretrained object classification network for feature extraction and training a model on an ecologically feasible manipulation

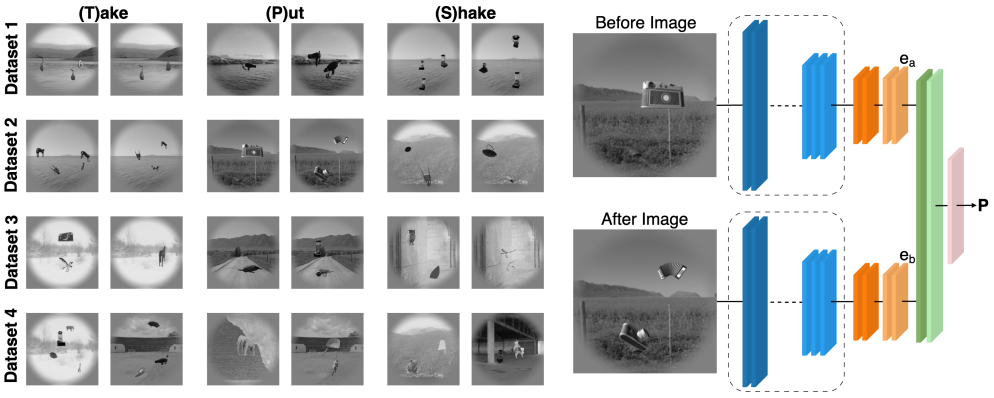


Fig. 1 Left: The stimuli datasets. Right: Our experimental setup, images before and after a manipulation (Take / Put / Shake) occurred are fed into a siamese network with shared weights consisting of a feature extraction model (in blue, comprised of the first consecutive blocks of an image classification network) and two blocks consisting of convolutions and dense layers (in orange). The output of the siamese branches $e_a, e_b \in \mathbb{R}^2$ are passed to a classification head (consisting of a convolution, linear, and softmax layers) which identifies which of the three manipulations occurred between the input image pairs.

recognition task with natural stimuli was sufficient for an ordinal one-dimensional numerosity embedding to emerge. Moreover, the ability to decode numerosity from the embedding was directly correlated with training task performance. Further analysis revealed that features extracted from intermediate, rather than late, object classification networks support better task performance and numerosity encoding. Furthermore, the numerosity encoding shared characteristics with human number-sense. Overall, these results suggest that primate like number-sense can evolve from commonplace interactions with the environment, and without need of explicit number labels. Moreover, this number sense is potentially driven by intermediate, rather than late and abstract, visual representations.

Our work is directly inspired by a growing body of literature which utilizes DNNs as cognitive models that provide insights into the functionality and hierarchical structure of cognitive representations [9–11].

2 Results

Following [7], we used a siamese DNNs trained to perform a 'manipulation recognition' task. Specifically, each model consists of two feature extraction branches and a classification head. For feature extraction, we used blocks from object classification models followed by two pairs of convolution and dense layers for further refining the representations. The classification head consists of a convolution layer followed by dense and softmax layers. The weights are shared across the two feature extraction branches, encoding the two input images into the same representational space (e_a, e_b in Figure 1).

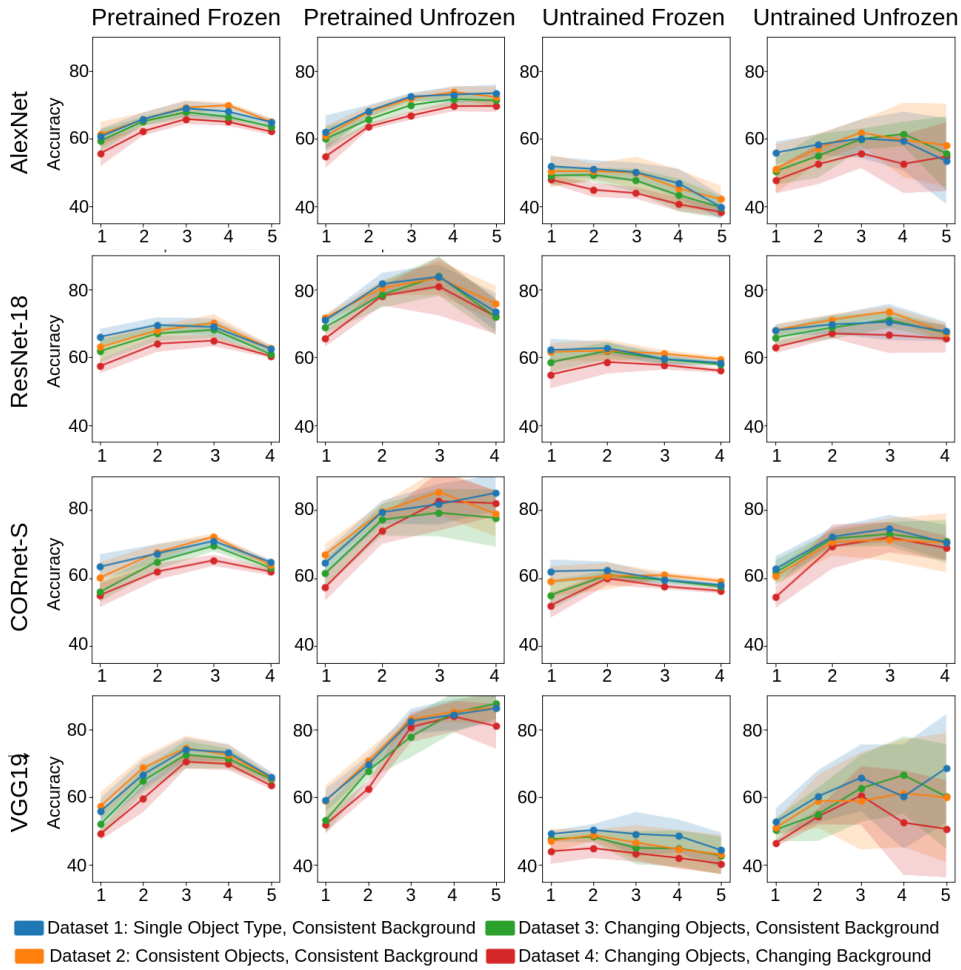


Fig. 2 Accuracy on the manipulation recognition task when using the first N blocks of various models (the x-axis). We used the AlexNet, ResNet-18, CORnet-S [12], and VGG-19 models with frozen/unfrozen, as well as, pretrained / randomly initialized weight and with various stimulus setups. All models were trained on images with up to four objects, and tested on images with up to eight objects.

2.1 Analyzing The Manipulation Recognition Performance of Various Visual Representation

We used four different classification models for feature extraction; AlexNet [13], ResNet-18 [14], CORnet-S [15], and VGG19 [16]. Each of these models consists of various blocks (five for AlexNet and VGG19, and four for ResNet-18 and CORnet-S) followed by a dense and softmax classification layers. Here we used the first consecutive N blocks (N ranging from 1 to the total number of blocks) for each of these four models for feature extraction. After passing the image through the feature extractor

two convolution and dense layers are used to further distill task relevant information from the feature-extractor representations. Finally, a classification head is used to perform the manipulation recognition task.

To reiterate, all models trained in this paper are siamese DNN where each branch consists of a feature-extractor and two randomly initialized convolution and linear layers (see figure 1 right).

Following [7], the first experiments used pre-trained and frozen feature-extractor weights. Replicating previous results, we find that the trained networks perform the task well, and even generalize to images with unseen numerosities (see the orange plot in the *Pretrained Frozen* column of Figure B1). Moreover, when considering results of feature-extractors consisting of different numbers of blocks a clear pattern emerges; Regardless of the model used for feature extraction, representations obtained from intermediate layers of the pretrained object recognition CNNs drove higher manipulation recognition results (see the orange plot in the *Pretrained Frozen* column of Figure B1). For statistical testing we used a one-tailed t-test between the accuracy achieved using representations from the first, third, and last block, validating that using intermediate, rather than early or late, representations led to significantly better accuracy for all *Pretrained Frozen* models.

These initial results could be accounted for by three hypothetical explanations: 1) The intermediate representation learned by object classification models are especially suitable for the manipulation recognition task. 2) Model architecture contributes significantly to performance, and an intermediate number of convolutional blocks is a 'Goldilocks' zone complexity wise 3) The model representations are approximating image statistics that, in intermediate layers, are confounded with numerosity or other task-relevant measures.

To test these hypotheses we used feature-extractor weights that are either unfrozen and / or untrained. As can be observed in Figure B1, models with randomly initialized weights (*Untrained Frozen* models) had lower performance relative to those pretrained to perform image classification, indicating that object recognition pre-training is useful. Moreover, unfrozen models outperformed models with frozen weights, indicating that mid-level representations of object recognition models are not simply better suited than high-level representations due to the level of granularity and abstraction they encode.

2.2 Task Performance and Numerosity Encoding

Previous work demonstrated that numerosity encoding emerged in models that were successfully trained to perform the manipulation task [7]. To validate this relationship between task performance and numerosity in our experiments we explore the encoding learned by the siamese network (e_a and e_b in Figure 1). First, we visually analyze the image embeddings for the best performing models, as can be observed in Figure 3 the image embeddings fall on a single-dimensional line that follows numeric order. To quantify how faithful this embedding is to the number label we calculated an *embedding score* by projecting the embeddings into the first PCA component and then training a regression to predict the image numerosity label. For an embedding score to be high the main principal component must encode numerosity in consecutive

order. The embedding scores of the pretrained frozen models with the standard stimuli peak in intermediate layers, resulting in a pattern similar to what was observed when plotting task performance (orange lines in the leftmost column of Figure B1). Moreover, manipulation recognition accuracy is highly correlated with embedding accuracy (correlation coefficient $R = 0.92$, see Figure 3), indicating that numerosity reliably emerged when training models using the Manipulation Recognition framework. In line with these results, when plotting the embedding scores for various models we observe similar patterns to those discussed in Section 2.1. In appendix

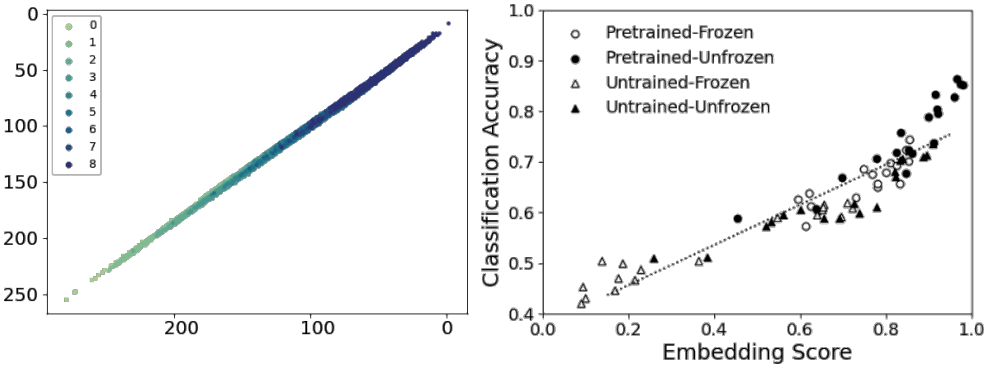


Fig. 3 Left: Embedding learned with a feature extraction using three VGG19 blocks with pretrained frozen weights, 32400 images with a number of objects between 0 and 8 are plotted. Right: correlation between performance on the manipulation recognition task and embedding accuracy. The two metrics have a high correlation of .92.

To ensure the numerosity encoding results do not reflect image confounds such as low-level statistics we extracted an array of measures (the Weibbull, Spatial Frequency Power, Spatial Frequency Magnitude) from 32400 validation images and correlated them with manipulation recognition task performance (Weibull: 0.02, spatial frequency power: 0.04, and spatial frequency magnitude: 0.41, the combined R value for all metrics was 0.51). Moreover, we repeated the experiments using a variety of stimulus setups (see Figure 1 left). If the models embed image statistics we would expect performance to drastically decrease between datasets 1 and 4. However, the decrease in performance was generally mild, and all experiments replicated the pattern of results of intermediate layers having the highest performance in pretrained frozen networks (see the first column of figure B1). Finally, we conducted multiple generalization experiments, in which a model was tested on a dataset it was not trained on. The model trained on Dataset 2 successfully generalized to Dataset 3 and Dataset 4 stimuli (See Appendix **??**), suggesting that model reliance on low level visual statistics, which are Dataset dependent, was limited.

2.3 Numerosity in Other Visual Tasks

Results presented in the previous sections indicate that visual representations at intermediate layers in object classification DNNs encode numerosity-relevant information.

This section explores whether DNNs trained to perform other visual tasks could similarly be used for feature extraction in the manipulation recognition framework. We used activations from 17 pre-trained deep neural network (DNN) models from the Taskonomy Task Bank [17]. The Task Bank consists of DNN that cover various computer vision tasks. We follow previous research demonstrating that the chosen subset of 17 tasks cluster into three categories based on the similarity of the features learned by the DNNs performing the tasks [18]. We refer to those three categories as 2D, 3D, and semantic tasks based on the following rationale; Models from the 2D task category process low-level visual features. Models from the 3D category process such mid-level features as surface normals and depth. Finally, models from the semantic category are trained to process high-level semantic features.

All Taskonomy models share weight initialization, and architecture, ensuring any difference in results is due to the influence of the task on the learned visual features. The models consist of an Encoder and Decoder, following previous research [10, 11] we analyze only the representations of the last encoder layer. All manipulation recognition models were trained on Dataset 1 images and with a *Pretrained Frozen* embedding network.

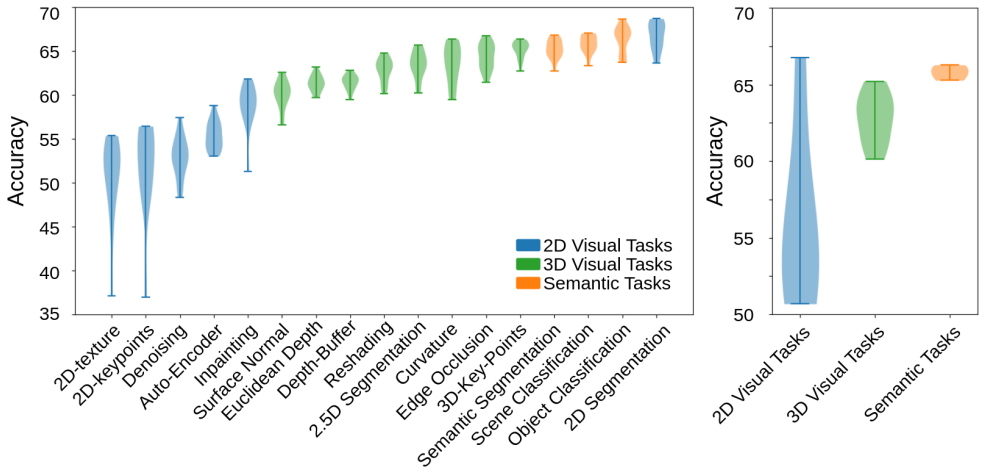


Fig. 4 Left: Mean performance for each and Taskonomy model, colors represent the type of visual task the feature extraction model was trained on. Right: The mean accuracy for each model group.

On average models that used semantic features had the highest performance (Figure 4 right). This seemingly contradicts results from previous sections as abstract and semantic representations usually emerge in late DNN layers. However, breaking down the results by task we observed that models that utilized representations learned by the *2 dimensional segmentation* feature extractor obtained the highest performance. As we later discuss, this aligns with intermediate visual representations being especially relevant for numerosity prediction.

2.4 Properties of the Numerosity Representations

Numerous modeling and primate studies found that numerosity encoding has various properties [4, 5]. Two widely researched properties are *Increasing Bandwidth* and *Positive Skewing*. Following these previous studies, we perform post-hoc analysis to investigate if the emerging numerosity representations also have similar properties.

The Increasing Bandwidth property refers to the fact that the numerosity label *response curves* increase for higher numerosities. Simply put, one can rapidly identify that a scene contains 3 objects, however it becomes likely to mislabel the number of objects for higher numerosities.

Positive Skewing similarly refers to the shape of the response curves. The positive skewing refers to the fact that these *response curves* are often positively skewed, which is manifested behaviourally by a higher likely-hood to exaggerate, rather than understate, the number of object in a scene.

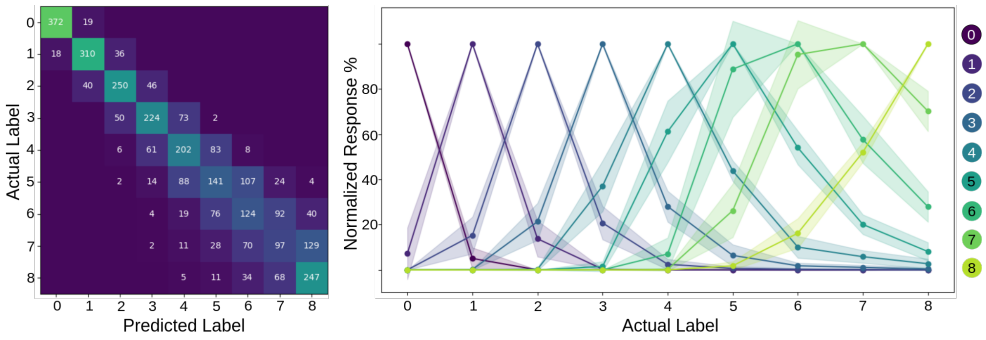


Fig. 5 Visualization for models VGG19-pretrained-frozen block 3 based model fine-tuned on dataset 1. Left: confusion matrix. Right: The response of various label prediction regressions normalized by the number of correctly labeled samples.

Following previous computational modeling work we first trained a regression to predict numerosity labels from our embedding [5]. The response curve for each label (the columns of the confusion matrix in Figure 5 left) were then fitted with a skewed normal distribution, which we used to calculate the variance and skewness for each response curve. A visualization of these results can be seen in Figure 5 right: the response curves after normalizing by the number of correctly labeled samples (hence the peak is always at 100%) are presented. This was done for all runs, enabling us to calculate average skewness (which was positive for all 9 response curve), similarly variances for all 10 runs were calculated and a page trend test was performed to verify the variance was indeed monotonically increasing with the numerosity label ($L = 2683$, $p < 0.001$). Similar results were achieved with similar models that used 3rd block representations from AlexNet, ResNet, and CorNet.

3 Discussion

Despite deep neural networks becoming increasingly common in tools in computational cognitive neuroscience, the debate regarding the significance and implications of computational modeling results is still ongoing. Here we exhaustively trained and tested over one thousand networks using an ecologically feasible task and stimuli, and reported several results that remained consistent across various training setups. We strongly believe that even a strict interpretation of our results as merely an indication of what is computationally possible - rather than as a plausible abstract model of the brain - can be of great interest to the research community focusing on studying primate number-sense.

First, we found that numerosity encoding emerges in models trained on an ecologically feasible *Manipulation Recognition* task and natural scene stimuli, replicating and extending previous work [7]. Moreover, analysis shows that *intermediate* visual representations learned from object classification networks are especially adaptable to both performing the manipulation recognition task and the emergence of numerosity encoding. This is especially interesting considering prior research demonstrated that representations learned by early, intermediate, and late layers of image classification DNNs correlate with early, intermediate, and late representations in the primate visual ventral stream [9, 19]. Considering this hierarchical correspondence, our results point towards intermediate visual representations playing a crucial role in the development of the primate visual number sense.

Further support for this argument can be found in the *taskonomy models* experiments (section 2.3). Out of all Taskonomy models, the two-dimensional segmentation task representations had the highest performance (Figure 4 left). These models were used in previous research as proxies for different brain representations [10, 11]. Specifically, while representations of three-dimensional and semantic visual tasks are correlated with advanced ventral and dorsal brain areas, representations of two-dimensional models correlate most with representations in earlier areas of the visual cortex.

Finally we also demonstrated that the learned numerosity embeddings have several attributes that were previously discovered in both primate brains and computational models. Though not directly relevant to our main result, these results further reinforce our hypothesis that these models, which were trained to perform a manipulation recognition task, learn robust numerosity encoding.

Overall, we believe that our results are evidence in support of an account of number-sense as relatively fundamental cognitive ability that emerges in relatively intermediate rather than (or in addition to) late brain areas that are more functionally semantic. We believe that further neuro-imaging research focusing on intermediate brain ares could be a promising venue of future work.

4 Methodology

All the code necessary to replicate our results was made publicly available[1].

[1] github.com/Arsu-Lab/Numerosity-Is-Driven-By-Intermediate-Visual-Representations

4.1 Stimuli

The stimuli consist of pairs of natural scenes that represent the state of an environment before and after a manipulation action occurred (a take action, a put action, and a shake). The stimuli consisted of 20 background images (10 natural and 10 artificial settings), 20 objects (10 animals and 10 tools), randomly positioned in one of 48 three rotations in three-dimensional space, and 20 levels of luminance. The size of the object was also varied to approximate the natural scene statistics [6]. The detailed data generation protocol can be found in Appendix A.

4.2 Datasets

The datasets are designed such that image pairs become increasingly irregular. The first dataset contains only a single type of object, with a pair containing the same background. The second dataset does not have restrictions on object types, however, the objects do not change beyond the addition and subtraction in Take and Put pairs. The third dataset takes this a step further, and all objects change between the image pairs. Finally in dataset four both the objects and backgrounds change between the image pairs.

4.3 Network Architecture

Our experimental setup followed previously proposed frameworks and methodologies [6, 7]. As can be observed in Figure 1, an image pair is fed into two branches of a siamese network, followed by a classification head. The siamese branches consist of a feature extraction model (comprised of a sub-part of an image classification DNN), followed by two CNN layers and one linear layer. We experiment with various setups of the feature extraction; we various sub-parts of AlexNet, ResNet-18, CORnet-S, and VGG19 with different combinations of pretrained/randomly initialized and frozen/unfrozen weights. The pretrained weights were obtained from the *torchvision* library, and all random weights (in the siamese branches and the classification head) were initialized with a Xavier initialization. The networks were trained using 162k pairs of images, epoch was batches of 30 180 pairs, and there were 30 epochs. Testing was done using 16200 pairs. The training images contained anywhere between 0 and 4 objects. However, our validation used images with up to 8 objects. When testing the models on data with up to 4 objects only the models' performance peaks were above 95%

4.4 Evaluation Metrics and Statistical Tests

To explore how the skewness and variance of the response curves change for different numbers. We first fitted each response curve with a skewed normal distribution. We then used the Page's trend test to explore if there are significant monotonic trends in variances and skewness across models.

Appendix A Stimulus Generation

Generating images involves several key steps: selecting and masking a background, choosing objects, and iteratively inserting these objects. Each step contributes to

creating a dataset that aligns with specific constraints like object selection and size range, varying with each dataset version.

1. **Selection of Background:** Our pool consists of 20 backgrounds —10 natural and 10 artificial settings. Refer to the supplementary material for details. Each background is a prerendered image of size 224×224 pixels. A spherical and centered perceptive field mask is applied to each background. This mask is designed to simulate the visual field in fMRI scan contexts, aiding in correlating our models with brain scan data if required in the future.
2. **Object Insertion:** Objects fall into two categories: tools and animals, with ten variations in each category. We use a range of prerendered images for each object, representing different angles and luminance levels. These properties are randomly selected across all datasets for each object. Additionally, we can adjust the object's apparent size in the image by scaling the prerendered image. During insertion, care is taken to avoid occlusion among objects within the perceptive field.

Algorithm 1 Algorithmic Outline for Generating Image Stimuli. It is important to note that the process for determining the *objectCount* and other attributes such as *Theta*, *Phi*, and similar, may vary and be substituted as required by different dataset variations.

```
Function GenerateStimuli(MaxNumerosity):
    background ← GetRandomBackground() maskedBackground ← MaskBackground(background) objectCount ← GetRandomObjectCount(MaxNumerosity)
    for i ← 1 to objectCount do
        luminance, theta, phi, objectType, size ← GetRandomParameters() object ← CreateObject(luminance, theta, phi, objectType, size)
        success ← False for attempt ← 1 to 1000 do
            insertionCoordinates ← GetInsertionCoordinates() success ← TryInsertImage(maskedBackground, object, insertionCoordinates)
            if success then
                break
            end
        end
        if not success then
            Print "Failed to insert image after 1000 attempts"
        end
    end
```

A.1 Object size variability

In order to decouple the numerosity from low-level feature properties, such as cumulative surface area, as best as possible, we vary the size of objects according to an exponential decay as introduced by Testolin et al. [6]:

$$a = (M - m) \times \exp(-b \cdot c) + m \tag{A1}$$

where a is the average object size. M is the maximum object size allowed (We use 100 pixels). m is the minimum object size allowed (We use 22 pixels). b is a constant factor (We use 0.5). c is the stimuli numerosity.

The average object size decreases as the numerosity increases. However, the object size variation is the greatest for small numerosities and decreases as the numerosity increases. This, in turn, makes it harder for the model to infer numerosity based on the cumulative surface area since cumulative surface areas of samples of different numerosities overlap.

Appendix B Embedding Scores Peak When Using Intermediate Model Feature

Similarly to accuracy, the embedding score also peaks when using features extracted from intermediate layers of the feature extraction model.

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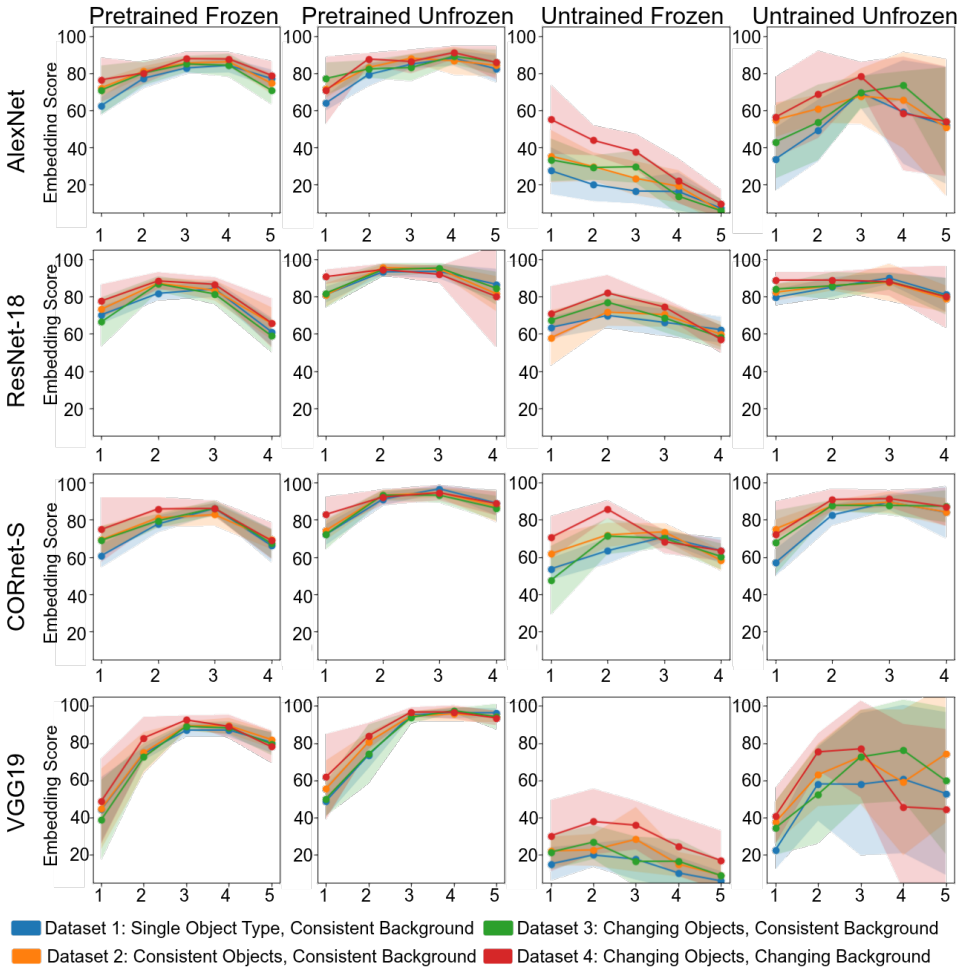


Fig. B1 Embedding Score when using the first N blocks of various models (the x-axis). We used the AlexNet, ResNet-18, CORnet-S [12], and VGG-19 models with frozen/unfrozen, as well as, pretrained / randomly initialized weight and with various stimulus setups. All models were trained on images with up to four objects, and tested on images with up to eight objects.

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